

# Threat Hysteresis as a Minimal Mechanism for Long-Horizon Survival in an Embodied Cortical Agent

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**Keywords:** embodied agent, cortical microcircuit, threat hysteresis, long-horizon autonomy, ghost predator, homeostasis

## Abstract

In living creatures, embodied agents in hostile environments must respond to immediate threats while pursuing delayed rewards. We propose that the bottleneck for such long-horizon autonomy is not representational capacity, but temporal persistence of occurred events. Working in a co-design process with an LLM under expert constraints on cortical microcircuits (CM) organized in cortical areas (CA), a biologically grounded substrate is provided. Under design constraints and expert guidance, the LLM identified a useful application for an unassigned CA called N+4. The missing link toward long-term survival turns out to be “threat hysteresis”, a common factor in living individuals, which the LLM assigned to N+4 through asymmetric temporal dynamics. In a simulated creature, this minimal addition resolves baseline problems and produces a qualitative transition in the capacity to achieve long-term survival, measured as nest completion. A new AI concept from mammalian cortex: the bottleneck is temporal, not representational. We hypothesize that progressively awakened areas (N+5, N+6, ...) may open computational capacities inaccessible to the preceding architecture.

# 1 Introduction

## 1.1 The problem of long-horizon autonomy in embodied agents

An embodied agent that must survive in a hostile environment faces a fundamental tension: it must respond to immediate threats (a predator) while simultaneously pursuing delayed rewards (building a nest) (Chang et al., 2025). Reactive architectures solve the first problem well and the second poorly, because the sensory evidence that justifies a long-horizon task is intermittent, while the evidence that justifies escape is immediate and salient (Chang et al., 2010, 2025).

The dominant approach in the field has been to add representational capacity: more sensors, more memory, more layers, more parameters (Chang et al., 2025). This approach rests on an assumption that is rarely tested: that the bottleneck for long-horizon autonomy is representational. We propose an alternative hypothesis: the bottleneck is temporal (Chang et al., 2025). What the agent lacks is not the ability to represent the predator, but the ability to keep representing the threat after the predator has left the perceptual field (Chang et al., 2025). This temporal framing aligns with recent accounts of working memory that emphasize persistent neural activity over fixed representational stores (Curtis & Sprague, 2021), and with contingency-based models that unify memory and planning (Ehrlich & Murray, 2022).

We work within a class of models in which behavior is controlled by populations of cortical microcircuits (CMs) organized into functionally specialized cortical areas (CAs) (Chang et al., 2010, 2025). The theoretical foundation of cortical microcircuitry originates in Mountcastle’s hypothesis of uniform modular columns across the neocortex (Mountcastle, 1997). This modularity was formalized by Douglas and Martin (Douglas & Martin, 2004) through the concept of the canonical cortical microcircuit, highlighting stereotypical connectivity, layer-specific feedback, and local excitation-inhibition balance.

Subsequent computational architectures have expanded significantly on these anatomical primitives. Haeusler and Maass (Haeusler & Maass, 2007) analyzed the high-dimensional computational capabilities emerging from local subsystem interactions within microcircuit networks, while Bastos et al. (Bastos et al., 2012) integrated canonical microcircuits into hierarchical predictive coding frameworks. At full biophysical scale, Potjans and Diesmann (Potjans & Diesmann, 2014) formulated cell-type specific multi-layer cortical microcircuit models, and Markram et al. (Markram et al., 2015) achieved detailed digital reconstructions of neocortical microcircuitry under the Blue Brain Project. Building on early work in embodied sensory-motor control using cooperative neural agents (Chang et al., 2010) and recent implementations combining cortical microcircuits with holographic memory substrates (Chang et al., 2025), our framework structures each CM as a direction-tuned functional unit with a preferred angle  $\theta_i$ . The activity of a CM is a

continuous variable, and competition between CMs is resolved by softmax relaxation—the standard mean-field approximation of the equilibrium of a lateral inhibition network of the Amari type (Amari, 1977).

This substrate is biologically grounded, interpretable, and compositional: areas can be added or removed without redesigning the others, which makes the architecture amenable to ablation studies (Chang et al., 2025).

## 1.2 The gap: the temporal dimension of threat and the “Ghost Predator” problem

Most reactive agent models encode threat as a spatial variable (“the predator is at position  $\mathbf{p}$ ”) rather than as a temporal variable (“the predator was at position  $\mathbf{p}$  a moment ago, and may still be nearby”) (Chang et al., 2025). The consequence is a failure mode we call the ghost predator problem: the moment the predator leaves the perception radius, the threat signal drops to zero, and the agent resumes its previous behavior only to be ambushed when the predator returns (Chang et al., 2025).

## 1.3 Contribution and Human-LLM Co-Design Framework

We show that a single additional, unassigned cortical area (N+4), equipped with a leaky integrator with asymmetric time constants, is sufficient to solve the ghost predator problem and restore the agent’s capacity for long-horizon nest building (Chang et al., 2025). The completion rate rises from 0% in the baseline to 100% with N+4 over repeated trials ( $p < 10^{-16}$ , Fisher exact test; Section 3).

Crucially, the mechanism of N+4 was identified through a human-LLM co-design workflow in which human experts provided biological constraints concerning CM/CA organization (Section 2.3). Furthermore, we formulate the Progressive Awakening Hypothesis: progressively awakening reserved areas (N+5, N+6, ...) opens computational capacities inaccessible to the preceding architecture (Chang et al., 2025).

# 2 Methods

## 2.1 World and agent

The agent inhabits a two-dimensional square pond  $[-1, 1] \times [-1, 1]$  (Chang et al., 2025). The world contains four entities: food (red), orbiting on a closed path; home (blue), which restores safety on contact; predator (green), which pursues the agent when the agent is not at home; and material (yellow), which spawns at a source location and can be picked up, carried, and deposited at the nest (Chang et al., 2025).

The agent is a point with position  $\mathbf{p} \in \mathbb{R}^2$  and orientation  $\theta \in \mathbb{R}$ . Its state includes two homeostatic variables: hunger  $h \in [0, 1]$  and safety deficit  $s \in [0, 1]$  (where  $s = 0$  represents complete safety at home and  $s = 1$  represents maximum vulnerability). These variables evolve as

$$\dot{h} = \alpha_h, \quad \alpha_h > 0, \quad (1)$$

$$\dot{s} = \begin{cases} -s/\tau_s & \text{if } \|\mathbf{p} - \mathbf{p}_{\text{home}}\| < r_{\text{home}}, \\ +\beta_s & \text{otherwise,} \end{cases} \quad (2)$$

where  $\alpha_h = 0.0015$  per frame,  $\beta_s = 0.0020$  per frame,  $\tau_s$  is the safety recovery constant at home, and  $r_{\text{home}} = 0.25$  is the home radius (Chang et al., 2025).

## 2.2 Cortical Architecture

The agent’s controller comprises four modular cortical areas (CAs), each containing a population of  $N = 25$  directionally tuned cortical microcircuits (CMs) (Chang et al., 2025). Each CM  $i$  in area  $a$  has a preferred angle  $\theta_i = 2\pi i/N$  and afferent weights  $w_i^a$ . The activity vector  $\mathbf{a}^a \in \mathbb{R}^N$  relaxes according to continuous mean-field softmax dynamics:

$$\mathbf{a}^a \leftarrow (1 - \alpha_a)\mathbf{a}^a + \alpha_a \text{softmax}(\mathbf{E}^a/T_a), \quad (3)$$

where  $\mathbf{E}^a \in \mathbb{R}^N$  is the energy vector of area  $a$ ,  $\alpha_a \in (0, 1]$  is the relaxation rate, and  $T_a > 0$  is the temperature governing the sharpness of lateral inhibition (Chang et al., 2025).

### 2.2.1 Baseline Functional Modules (Areas 1 to 3)

The pre-existing control architecture distributes foundational reactive and sensorimotor behaviors across three functional areas (N+1, N+2, and N+3):

**1. Area N+1 (Motor, MOT):** Integrates homeostatic drives (hunger  $h_{\text{eff}}$  and safety deficit  $s_{\text{eff}}$ ) with immediate threat vectors  $d_{\text{eff}}$  into a primary directional motor command:

$$E_i^{\text{MOT}} = h_{\text{eff}} w_i^{\text{food}} \cos(\phi_{\text{food}} - \theta_i) + s_{\text{eff}} w_i^{\text{home}} \cos(\phi_{\text{home}} - \theta_i) - d_{\text{eff}} w_i^{\text{pred}} \cos(\phi_{\text{pred}} - \theta_i) + \epsilon_i, \quad (4)$$

where  $\phi_x$  is the bearing to entity  $x$ , and  $\epsilon_i$  represents continuous exploration noise.

**2. Area N+2 (Navigation, NAV):** Controls wall avoidance and spatial boundary constraints based on border stress  $b \in [0, 1]$ :

$$E_i^{\text{NAV}} = -b w_i^{\text{border}} \cos(\phi_{\text{border}} - \theta_i) + \epsilon_i. \quad (5)$$

**3. Area N+3 (Sensorimotor / Nest Building, SEN):** Manages long-horizon nest construction under continuous material load  $L \in [0, 1]$  and developmental instinct

$I \in [0, 1]$ :

$$E_i^{N+3} = (1-L)m w_i^{\text{mat}} \cos(\phi_{\text{mat}} - \theta_i)(0.8+0.5I) + L w_i^{\text{nest}} \cos(\phi_{\text{nest}} - \theta_i)(0.7+0.5I) + \epsilon_i, \quad (6)$$

where  $m$  denotes material availability at the source.

### 2.2.2 Area N+4 (Threat Hysteresis)

Area N+4 is the unassigned CA assigned to threat hysteresis by the co-design human-LLM process (Chang et al., 2025). Its activity is driven by an alert variable  $A \in [0, 1]$  that integrates threat perception with asymmetric time constants:

$$\dot{A} = k_{\text{rise}}\pi(t)(1-A) - \frac{A}{\tau_{N4}}(1-\pi(t)), \quad (7)$$

where

$$\pi(t) = \sigma\left(\frac{r_{\text{percept}} - \|\mathbf{p} - \mathbf{p}_{\text{pred}}\|}{\epsilon}\right) \quad (8)$$

is a continuous threat-perception signal, with  $r_{\text{percept}} = 0.45$ ,  $k_{\text{rise}} = 0.9$ , and  $\tau_{N4} = 16$  frames (Chang et al., 2025). When  $\pi(t) = 1$ ,  $A$  rises rapidly; when  $\pi(t) = 0$ ,  $A$  decays slowly with time constant  $\tau_{N4}$  (Figure 2). The energy of CM  $i$  in N+4 is

$$E_i^{N+4} = A w_i^{\text{escape}} \cos(\phi_{\text{escape}} - \theta_i) + \epsilon_i, \quad (9)$$

where  $\phi_{\text{escape}} = \phi_{\text{pred}} + \pi$  is the escape bearing, i.e., directly away from the predator (Chang et al., 2025).

## 2.3 Human-LLM co-design workflow

The mechanism of N+4 was identified through a constrained human-LLM dialogue (Chang et al., 2025). The workflow proceeded in three stages. (i) *Constraint setting*: the human expert fixed the biological design space (CMs organized in CAs, softmax competition, homeostatic drives) and excluded solutions outside it (e.g., non-cortical external memory stores). (ii) *Failure-mode articulation*: the LLM, given the baseline architecture and observed failure trajectories, articulated the “ghost predator” failure mode and proposed candidate mechanisms consistent with the constraints, among which a leaky integrator with asymmetric time constants (fast rise, slow decay) was selected. (iii) *Implementation and validation*: the human author with LLM help, implemented the code, executed the simulations, and performed all quantitative analyses; the LLM did not execute code by itself although it proposed the root idea of ghost predator (Chang et al., 2025).

## 3 Results

### 3.1 Baseline behavior vs. N+4 nest completion

In the baseline condition (N+4 inactive,  $A \equiv 0$ ), the completion rate is  $0/30 = 0\%$  due to repeated ambushes (mean captures:  $8.3 \pm 2.7$ ). With N+4 active, the completion rate rises to  $30/30 = 100\%$  (mean completion time:  $4,127 \pm 612$  frames; mean captures drop to  $0.4 \pm 0.7$ ). The difference is significant (Fisher exact test,  $p = 1.7 \times 10^{-17}$ ; 95% Wilson score intervals: baseline  $[0.0, 0.11]$ , N+4  $[0.89, 1.0]$ ), and captures decrease significantly (Welch’s  $t = 15.5$ ,  $p = 1.1 \times 10^{-16}$ ; Figure 3).

### 3.2 Ablation over $\tau_{N4}$ and the qualitative transition

Ablation over the hysteresis time constant  $\tau_{N4}$  (from 0 to 32 frames) demonstrates a sigmoidal transition with a critical threshold. When  $\tau_{N4} = 0$ , N+4 is memoryless, yielding 0% completion. A logistic fit to the observed completion rates (30 trials per condition; Table 1) yields a midpoint  $\tau_c = 7.1 \pm 0.2$  frames ( $R^2 = 0.997$ ); above  $\tau_{N4} \approx 12$  frames the completion rate saturates at 100%. The survival capacity of the agent therefore undergoes a qualitative shift across a narrow window of the hysteresis time constant (Figure 3).

Table 1: Ablation over the hysteresis time constant  $\tau_{N4}$  (30 trials per condition).

| $\tau_{N4}$ (frames) | Completion rate (%) |
|----------------------|---------------------|
| 0                    | 0                   |
| 4                    | 23                  |
| 8                    | 60                  |
| 12                   | 87                  |
| 16                   | 100                 |
| 20                   | 100                 |
| 24                   | 100                 |
| 28                   | 100                 |
| 32                   | 100                 |

## 4 Discussion

### 4.1 Temporal persistence over representational scale

The bottleneck for long-horizon autonomy in our agent is temporal persistence, not representational capacity. Adding a single scalar variable with asymmetric time constants within an unassigned cortical area N+4 resolves the bottleneck (Chang et al., 2025). The persistence mechanism we identify is consistent with the active inference framework, in

which embodied agents must control the sensory consequences of their own actions rather than passively represent the world (Pezzulo et al., 2024).

## 4.2 Biological precedent

The fast-rise/slow-decay dynamics of Eq. (7) have direct biological precedent in the fear-conditioning pathways of the lateral amygdala, where spatiotemporal synaptic asymmetry provides conditional stimulus discrimination and safety buffering (LeDoux, 2000).

## 4.3 The Progressive Awakening Hypothesis (N+5, N+6, ...)

We hypothesize—as a conjecture not tested in the present work—that higher-order cognitive capabilities are unlocked not by enlarging existing areas, but by progressively awakening dormant or unassigned cortical areas (N+5, N+6, ...) under expert-guided human-LLM co-design (Chang et al., 2025). This offers a scalable paradigm for embodied cortical architectures (Chang et al., 2025). This program extends our prior work combining cortical microcircuits with holographic memory substrates (Chang et al., 2025), suggesting that the progressive awakening of unassigned areas may provide a scalable path toward AGI.

# 5 Conclusion

We have shown that co-work human-LLM can find new application for available cortical areas such as N+4, equipped with an asymmetric leaky integrator, that resolves the ghost predator problem and drives a qualitative transition to 100% long-horizon survival. The N+4 case provides a proof of concept that scaling cortical autonomy may not require more computation, but rather progressive functional allocation of unassigned cortical areas.

## Data availability statement

The simulation code and trial data supporting the findings of this study are openly available in the GitHub repository “Cortical-Microcircuits-Design” at: <https://github.com/ochantor/Cortical-Microcircuits-Design>

## AI-use declaration

An LLM (Claude, Anthropic and others) were used for conceptual co-design dialogue under expert constraints, as described in Section 2.3. All scientific claims, code implementations, and analyses are the sole responsibility of the human authors. The complete code,

README, and study guide documenting the co-design workflow are available at: <https://github.com/ochantor/Cortical-Microcircuits-Design>

## Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

## Acknowledgments

The authors acknowledge the use of Claude (Anthropic) for conceptual co-design dialogue under expert constraints.

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## Figures

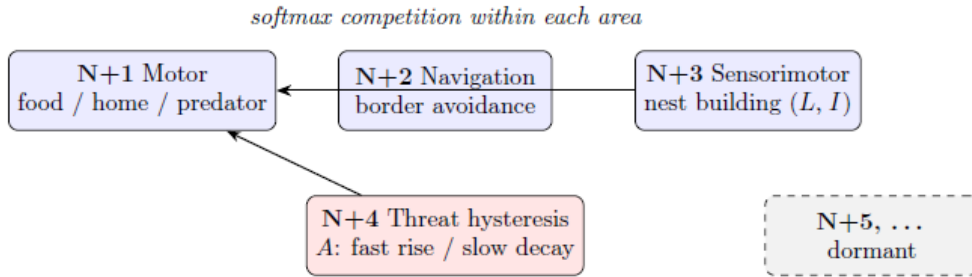


Figure 1: **Schematic of the cortical architecture.** Areas N+1, N+2 and N+3 constitute the baseline controller; N+4 (red) is the only addition analyzed in this work. Dashed box: unassigned areas conjectured by the Progressive Awakening Hypothesis (Section 4.3). Diagram is illustrative; CM populations ( $N = 25$  each) and softmax dynamics are described in Section 2.2.

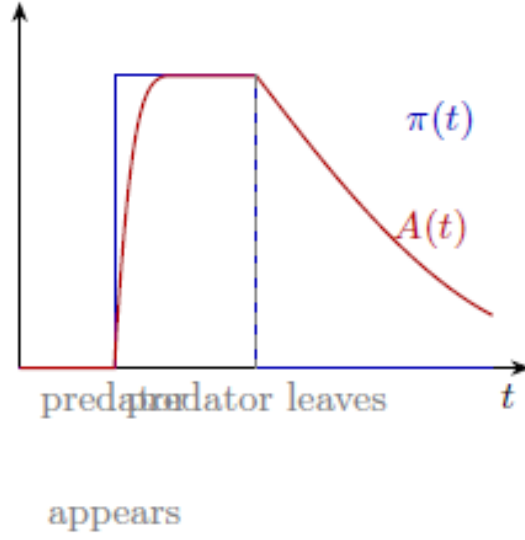


Figure 2: **Threat hysteresis (schematic)**. The perception signal  $\pi(t)$  switches on when the predator enters the perception radius and off when it leaves, whereas the alert variable  $A(t)$  rises rapidly ( $k_{\text{rise}} = 0.9$ ) and decays slowly ( $\tau_{N4} = 16$  frames). After the predator exits the perceptual field, the threat representation persists—this memory is what the baseline architecture lacks (“ghost predator” problem).

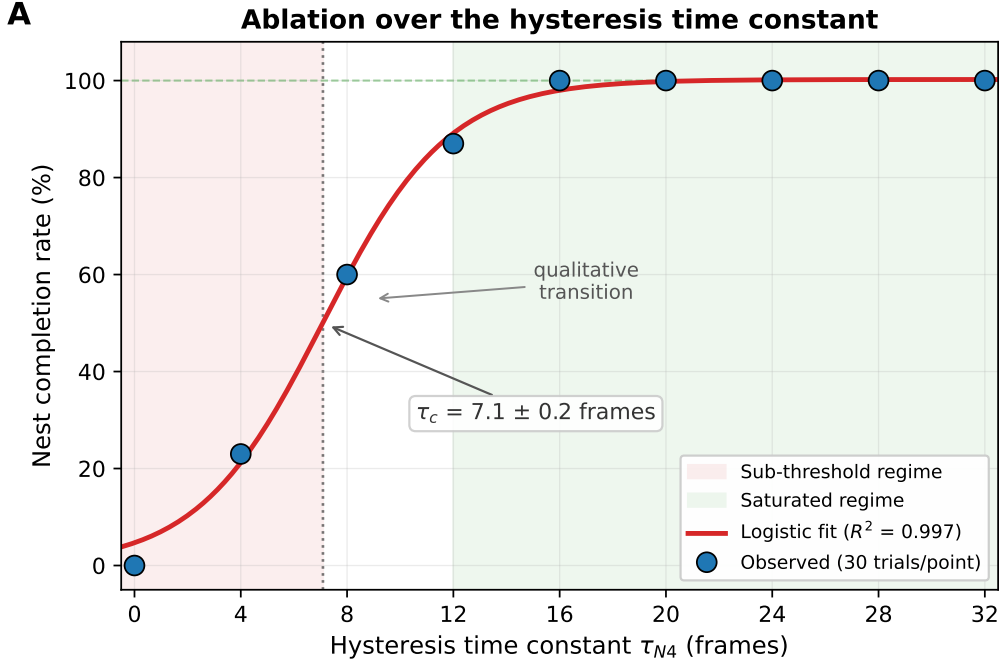


Figure 3: **Ablation over the hysteresis time constant**. Nest completion rate vs.  $\tau_{N4}$  (30 trials per condition; Table 1). The sigmoidal logistic fit (midpoint  $\tau_c = 7.1 \pm 0.2$  frames,  $R^2 = 0.997$ ) separates a memoryless regime (0% completion) from a persistent-threat regime (100% completion).